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EAS 4300 HW #6 Question 3

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% Clear workspace, close all figures, and clear command window
clear all; close all; clc;

Constants

```
f_st = 0.06; % Stoichiometric fuel-air ratio
T0_4_max = 2600; % Maximum combustor temperature (K)
delta_H_c = 43000; % Heat of combustion (kJ/kg)
k = [1.4 1.33]; % Specific heat ratios (inlet: 1.4,
combustor: 1.33)
R = 0.287; % Gas constant (kJ/kg·K)
delta = 1000; % Number of points for Mach range

% Ambient conditions at 10,000 meters
T_a = 223.252; % Ambient temperature (K)
P_a = 2.65e4; % Ambient pressure (Pa)

% Specific heat at constant pressure
Cp_air = (k(1)/(k(1)-1)) * R; % Inlet air, kJ/kg·K
Cp = (k(2)/(k(2)-1)) * R; % Combustor products, kJ/kg·K

% Preallocation of variables
M_flight = linspace(1, 6, delta);
n = length(M_flight);
P0_a = zeros(n,1);
Me_t = zeros(n,1);
A_exit_A_throat = zeros(n,1); % Renamed for clarity
T0_a = zeros(n,1);
f = zeros(n,1);
```

```

T04 = zeros(n,1);
T6 = zeros(n,1);
u_e = zeros(n,1);
u = zeros(n,1);
I = zeros(n,1); % Specific thrust (m/s)
TSFC = zeros(n,1); % Thrust specific fuel consumption (kg/N.s)
eta_th = zeros(n,1);
eta_p = zeros(n,1);
eta_0 = zeros(n,1);

```

Ramjet Performance, $M_{\text{flight}} = 1$ to 6

```

for i = 1:n
    % Stagnation pressure at inlet
    P0_a(i) = P_a * (1 + ((k(1) - 1)/2) * M_flight(i)^2)^(k(1) / (k(1) - 1));

    % Flight velocity
    u(i) = M_flight(i) * sqrt(k(1) * R * 1000 * T_a); % m/s

    % Exit Mach number: ideal expansion back to P_a with P0_e = P0_a
    Me_t(i) = sqrt((2/(k(2)-1)) * ((P0_a(i)/P_a)^((k(2)-1)/k(2)) - 1));

    % Area ratio (A_exit / A_throat)
    A_exit_A_throat(i) = (1/Me_t(i)) * ((2/(k(2)+1)) * (1 + ((k(2)-1)/2) *
    Me_t(i)^2))^(k(2)+1)/(2*(k(2)-1)));

    % Stagnation temperature at inlet
    T0_a(i) = T_a * (1 + ((k(1)-1)/2) * M_flight(i)^2);

    % Burner energy balance: Cp_air*T0_a + f*delta_H_c = (1 + f)*Cp*T04
    f(i) = (Cp * T0_4_max - Cp_air * T0_a(i)) / (delta_H_c - Cp * T0_4_max);

    % If reaching T0_4_max would need more than stoichiometric fuel, burn
    % at f_st and find the lower T04 it reaches
    if (f(i) > f_st)
        f(i) = f_st;
        T04(i) = (Cp_air * T0_a(i) + f_st * delta_H_c) / ((1 + f_st) * Cp);
    else
        T04(i) = T0_4_max;
    end

    % Exit temperature
    T6(i) = T04(i) / (1 + ((k(2)-1)/2) * Me_t(i)^2);

    % Exit velocity
    u_e(i) = Me_t(i) * sqrt(k(2) * R * 1000 * T6(i)); % m/s

    % Specific thrust
    I(i) = (1 + f(i)) * u_e(i) - u(i); % m/s

    % Thrust specific fuel consumption
    TSFC(i) = f(i) / I(i); % kg/(N.s)

```

```

    % Efficiencies
    eta_p(i) = (I(i) * u(i)) / (((1 + f(i)) * (u_e(i)^2) / 2) - (u(i)^2) / 2);
    eta_th(i) = ((1 + f(i)) * u_e(i)^2 - u(i)^2) / (2 * f(i) * delta_H_c *
1000);
    eta_0(i) = eta_th(i) * eta_p(i);
end

% Ensure Mach numbers are a column vector for the data table
M_flight = M_flight(:);

```

Figure 1: Specific Thrust vs. Flight Mach Number

```

figure(1)
plot(M_flight, I, 'LineWidth', 1.5)
xticks(1:6)
ylim([400 1200])
xlabel('Flight Mach Number (M_{flight})')
ylabel('Specific Thrust (m/s)')
title('Specific Thrust vs. Flight Mach Number')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure1.pdf')

```

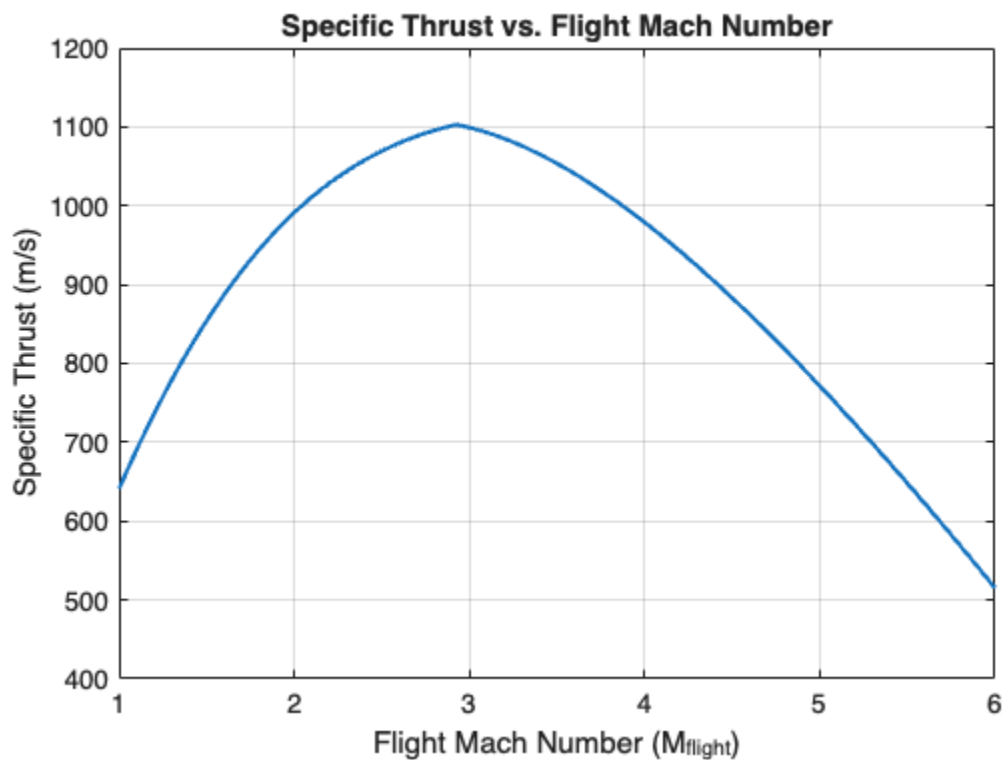


Figure 2: TSFC vs. Flight Mach Number

```
figure(2)
plot(M_flight, TSFC, 'LineWidth', 1.5)
xticks(1:6)
ax = gca;
ax.YAxis.Exponent = 0;
ylim([0.00004 0.0001])
xlabel('Flight Mach Number ( $M_{flight}$ )')
ylabel('TSFC (kg/N·s)')
title('Thrust Specific Fuel Consumption vs. Flight Mach Number')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure2.pdf')
```

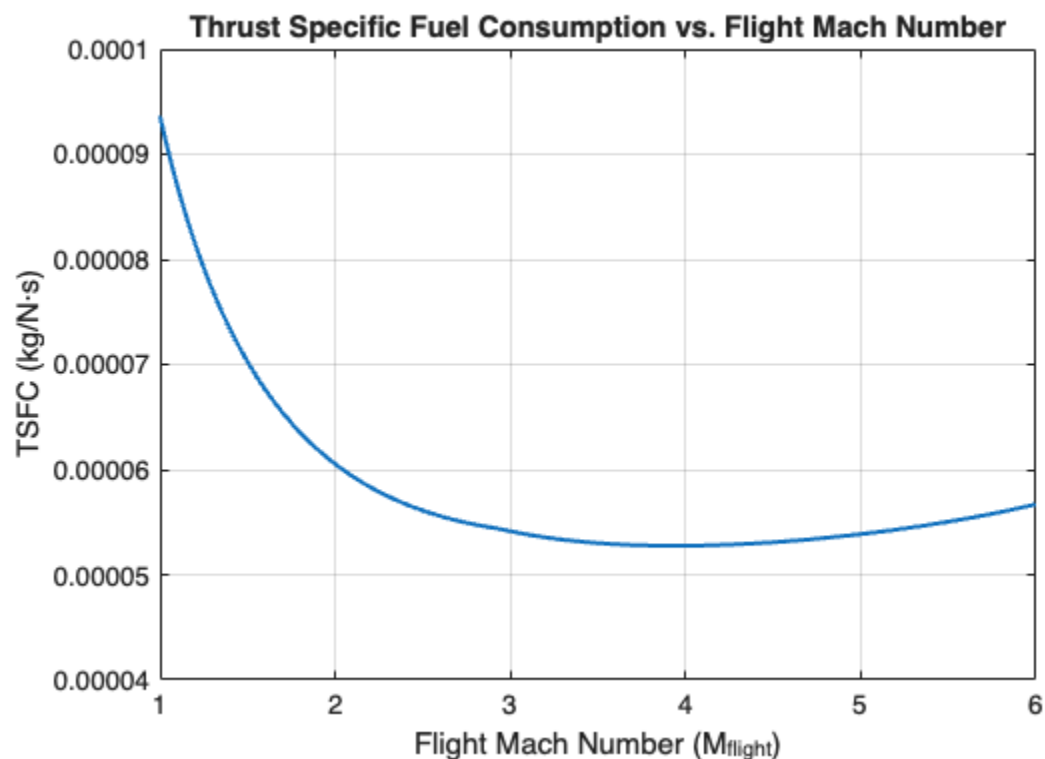


Figure 3: Combustor Exit Temperature vs. Flight Mach Number

```
figure(3)
plot(M_flight, T04, 'LineWidth', 1.5)
xticks(1:6)
ylim([2300 2700])
xlabel('Flight Mach Number ( $M_{flight}$ )')
ylabel('Combustor Exit Temperature ( $T_{0_4}$ , K)')
title('Combustor Exit Temperature vs. Flight Mach Number')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure3.pdf')
```

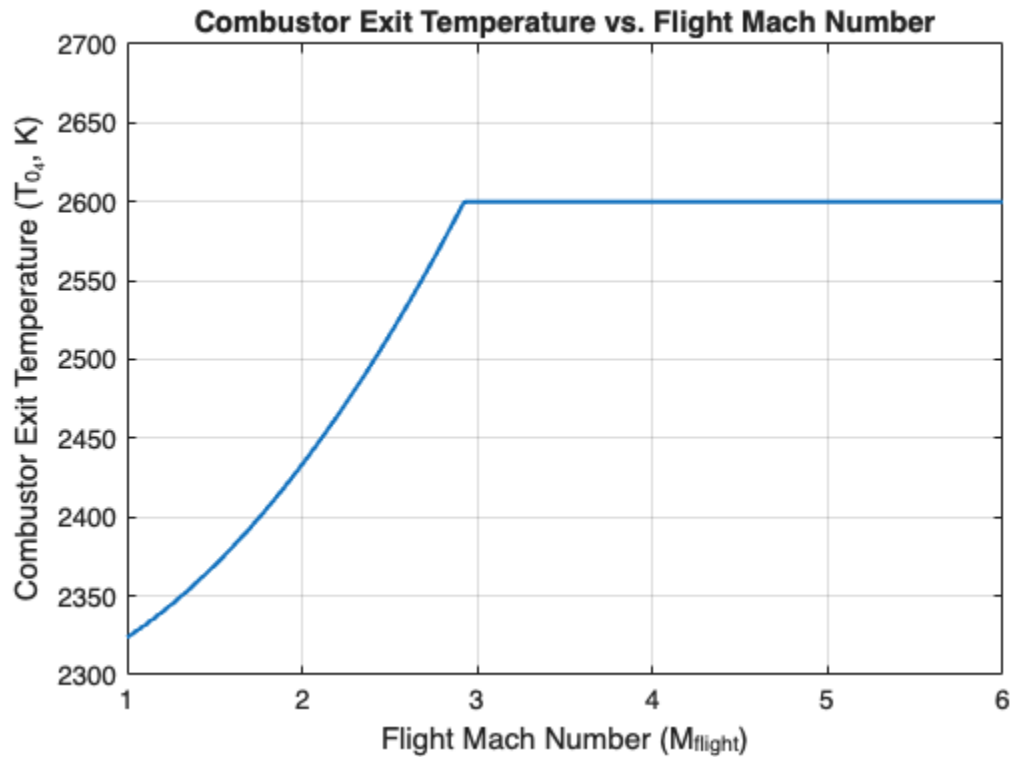


Figure 4: Nozzle Area Ratio vs. Flight Mach Number

```
figure(4)
plot(M_flight, A_exit_A_throat, 'LineWidth', 1.5)
xticks(1:6)
xlabel('Flight Mach Number ( $M_{flight}$ )')
ylabel('Area Ratio ( $A_{exit}/A_{throat}$ )')
title('Area Ratio vs. Flight Mach Number')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure4.pdf')
```

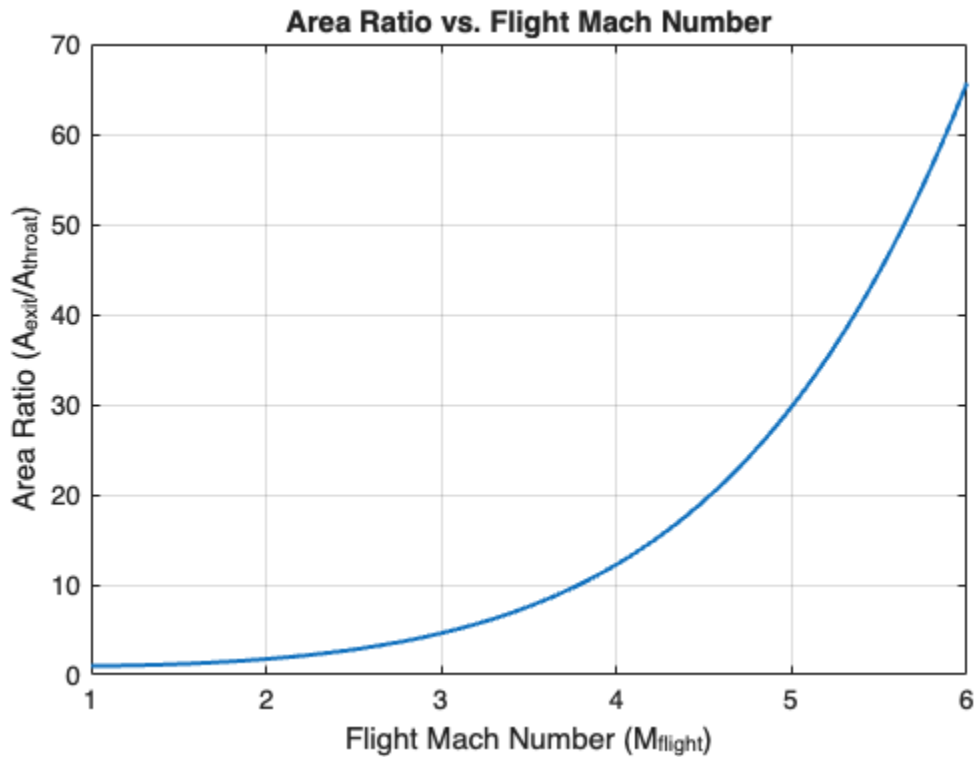


Figure 5: Efficiencies vs. Flight Mach Number

```
figure(5)
plot(M_flight, eta_0, 'b-', 'LineWidth', 1.5, 'DisplayName', '\eta_{0}')
hold on
plot(M_flight, eta_th, 'r-', 'LineWidth', 1.5, 'DisplayName', '\eta_{th}')
plot(M_flight, eta_p, 'g-', 'LineWidth', 1.5, 'DisplayName', '\eta_{p}')
hold off
xlabel('Flight Mach Number ( $M_{flight}$ )')
ylabel('Efficiency')
title('Efficiencies vs. Flight Mach Number')
legend('Location', 'best')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure5.pdf')
```

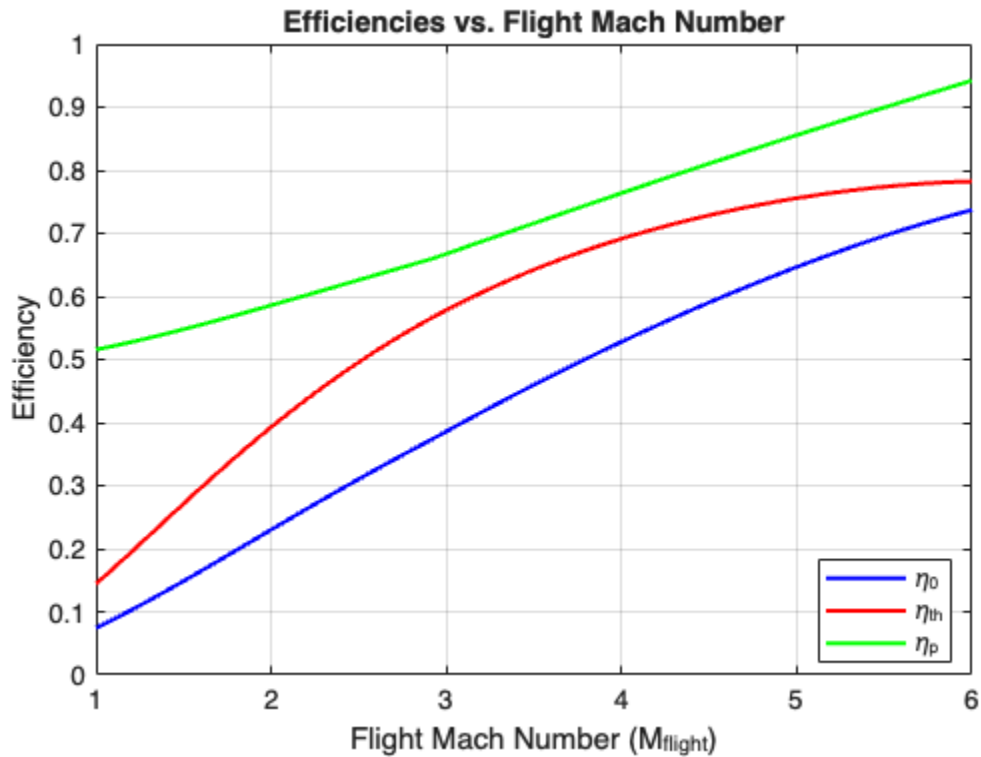
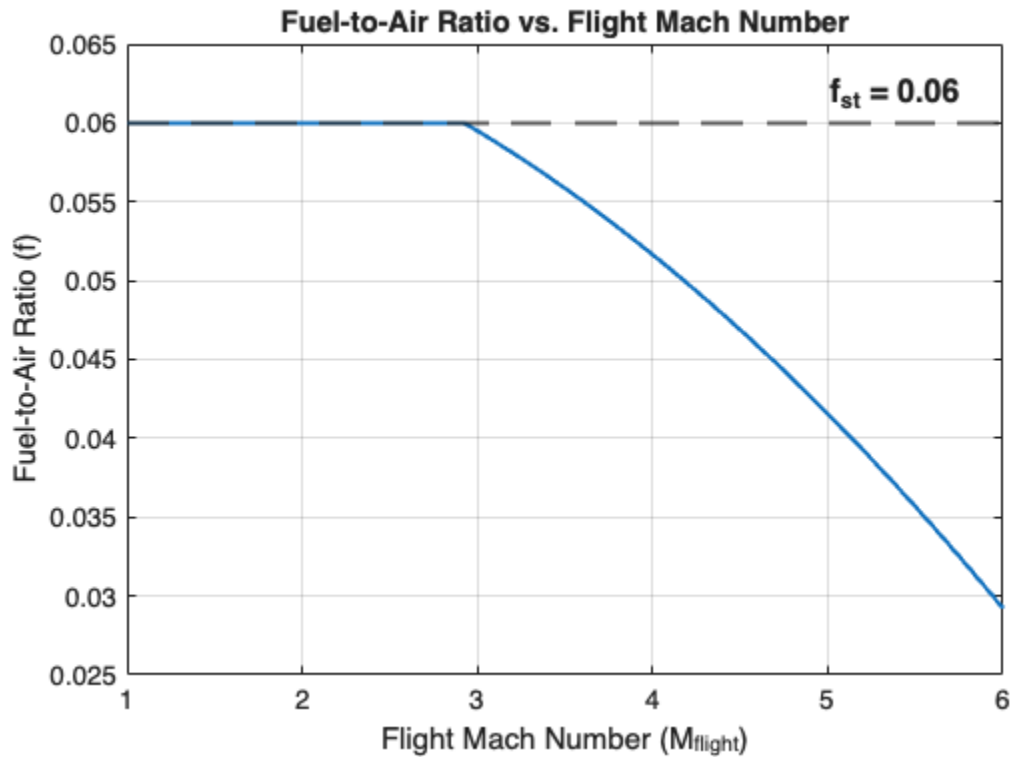


Figure 6: Fuel-to-Air Ratio vs. Flight Mach Number

```
figure(6)
plot(M_flight, f, 'LineWidth', 1.5)
xticks(1:6)
ylim([0.025 0.065])
xlabel('Flight Mach Number ( $M_{\text{flight}}$ )')
ylabel('Fuel-to-Air Ratio (f)')
yline(f_st, '--', 'LineWidth', 1.5)
text(5, 0.062, 'f_{st} = 0.06', 'FontWeight', 'bold', 'FontSize', 12)
title('Fuel-to-Air Ratio vs. Flight Mach Number')
grid on
print('-dpdf', '-bestfit', 'HW6_Q3_Figure6.pdf')
```



Save Results

```
data = table(M_flight, Me_t, u, u_e, f, I, TSFC, T04, A_exit_A_throat,  
eta_th, eta_p, eta_0, ...  
    'VariableNames', {'M_flight', 'Me_t', 'u', 'u_e', 'f', 'Specific_Thrust',  
    'TSFC', 'T04', 'A_exit_A_throat', 'eta_th', 'eta_p', 'eta_0'});
```

```
% Save data to a file  
writetable(data, 'hw6_q3_results.csv');
```

```
disp('Calculation and plotting complete. Results saved to  
hw6_q3_results.csv.');
```

Calculation and plotting complete. Results saved to hw6_q3_results.csv.

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